Streichwurst 0.1.0 – User Manual

String machine with solo voices and ensemble effects for the Electro-Smith Seed3 Desktop Development Kit · Test build 1 · 1 November 2026 · built from `main` @ `0a29966`

Streichwurst is a polyphonic string machine in the spirit of the Waldorf Streichfett: a fully polyphonic string section, eight solo voices, and an effects section with Animate, Ensemble, Phaser and Reverb. It is its own implementation built from documented research; it is not a copy of the original and does not claim to sound identical. The string section and the ensemble follow the documented structure of the original most closely; the soli, Animate, Phaser and Reverb are own designs.

Image	Purpose	Size	SHA-256
Streichwurst-v0.1.0.bin	The instrument: strings, soli, FX, Mod on Daisy matrix	239 020 B	`f244ae6bf53d452b…21681ca75e9`
StreichwurstFx-v0.1.0.bin	The FX section alone, for external audio (line input)	164 508 B	`1a0f4c96310af322…aa55185479f9b6`

> **Status:** host tests, sanitizers, the control app self-test and both ARM builds passed. Sound, controls, MIDI, CPU load and the comparison with the original on real hardware are **NOT MEASURED**. Start with a low listening level.

1. Installing the firmware

- Compare the checksum with `SHA256SUMS.txt` before flashing.
- First time or recovery: hold **BOOT**, press **RESET**, release BOOT, then flash over DFU (Daisy Web Programmer or `make PROJECT=streichwurst program-dfu`).
- Without the Seed buttons: hold **S9 + S16** (and no other key) for **5 s**, or use **Firmware-Update …** in the control app. All 16 LEDs light up and the bootloader waits for DFU without a time limit.
- Recommended order: first test the FX image alone, then the instrument (“Modular FX first”).

2. Connections

Connector	Use
Audio OUT L/R, headphones	Stereo output, 48 kHz
USB J7	USB-MIDI: notes, CCs, the control app (SQ protocol and Mod on Daisy on the same port), firmware update
MIDI DIN in	Notes and CCs (no SysEx)
CV IN 1/2, Gate IN 1/2	Sources of the modulation matrix (CV 1/2; gates clock the LFOs)
Audio IN L/R	Only used by the FX image (left alone = mono to both sides)

3. How it sounds and is built

```
Strings  -> Animate -> Ensemble -> Phaser -> stereo  \
Soli (bypass the string effects) ------------------>  Balance -> Reverb -> Master -> limiter -> OUT
```

- **Strings:** all 128 notes are available at the same time. Twelve free-running clocks (one per pitch class) are divided down into all octaves, like on classic string machines. **Registration** morphs from a dark, full sound to a thin, bright one; **Octaves** selects the base footage, both, or the upper octave. **Crescendo** and **Release** are the swell and fade times (about 0.6 ms to 4.4 s).
- **Soli:** eight voices with a Tone morph (hollow square → saw → narrow pulse, darker → brighter), attack, release, sustain level, three sustain modes and tremolo. In **Split** mode the keys below the split key play the strings and the keys from the split key up play the soli; in **Layer** mode every key plays both.
- **Ensemble:** three modes with 3, 5 and 3 modulated delay taps and three LFOs (0.32, 0.81 and 7.77 Hz). Switching the mode or the ensemble on/off cross-fades in 20 ms. As in the original, the ensemble makes the whole output louder or quieter depending on the mode (mode 1 is the densest).
- **Animate** slowly moves the registration and adds a wandering band emphasis; **Phaser** sweeps the ensemble sum; **Reverb** works on the whole mix including the soli.
- **Balance** in the middle plays both sections at full level; turning it to one side fades the other out.

4. Controls on the kit

The eight pots have three pages. **S9** switches Main → Solo → FX; S9–S11 show the active page (a short blink after a change). After a page change, a preset, or a change from MIDI or the app, a pot does nothing until you turn it across the current value (**soft takeover**), so values never jump.

Pot	Main page	Solo page	FX page
VR1	Registration	Solo Tone	Animate
VR2	Crescendo	Solo Attack	Animate rate
VR3	Release (strings)	Solo Release	Phaser
VR4	Balance	Tremolo depth	Phaser rate
VR5	Animate	Tremolo rate	Reverb
VR6	Phaser	Solo Sustain	Reverb decay
VR7	Reverb	Split key	Master
VR8	Master	Master	Balance

Key	Function	LED
S1 – S8	Play C4 D4 E4 F4 G4 A4 B4 C5 on the MIDI channel	lit while held
S9	Next pot page (on release)	S9/S10/S11 = Main/Solo/FX page
S10	Octaves: Base → Both → Upper	
S11	Ensemble mode 0 → 1 → 2	
S12	Ensemble on/off	lit = on
S13	Solo sustain mode: Decay to level → Hold → Percussive	off / dim / bright
S14	Layer / Split	lit = Split
S15	Next RAM preset (0–3)	blinks preset number + 1 times
S16	Panic: all sound off with a 2 ms fade (on release)	lit during the fade
S9 + S16	Held alone for 5 s: firmware update (bootloader). Neither page change nor panic.	all LEDs

Sustain modes: *Decay to level* falls to the Solo Sustain level (80 ms); *Hold* stays at full level like an organ; *Percussive* fades out even while held (0.05–3.2 s, set with Solo Sustain).

RAM presets (own presets, nothing is stored in flash): 0 defaults · 1 dark ensemble strings · 2 bright strings with both octaves, phaser and Animate · 3 split: strings below C4, solo lead above. A preset keeps the MIDI channel, Master and Balance.

5. MIDI

- Notes, velocity, pitch bend (±2 semitones by default), sustain pedal (CC64), All Notes Off (CC123) and All Sound Off (CC120). Pitch bend moves strings and soli together.
- **MIDI channel:** omni by default; set 1–16 in the app (Global page). Changing it releases held notes.
- The strings ignore velocity by default (Velocity (strings) = 0 %); the soli follow velocity fully.
- CC1 (mod wheel) and aftertouch belong to the modulation matrix.

CC	Parameter	CC	Parameter
CC 7	Master	**CC 31**	Layer / Split
CC 8	Balance strings/solo	**CC 85**	Split key (MIDI note)
CC 20	Registration	**CC 86**	Velocity (strings)
CC 21	Octaves	**CC 87**	Velocity (solo)
CC 22	Crescendo	**CC 102**	Animate
CC 23	Release (strings)	**CC 103**	Animate rate
CC 24	Solo Tone	**CC 104**	Ensemble on/off
CC 25	Solo Attack	**CC 105**	Ensemble mode
CC 26	Solo Release	**CC 106**	Phaser
CC 27	Solo sustain mode	**CC 107**	Phaser rate
CC 28	Solo Sustain	**CC 108**	Reverb
CC 29	Tremolo depth	**CC 109**	Reverb decay
CC 30	Tremolo rate		

Continuous parameters use the full CC range; switches use equal zones (on from 64).

6. Control app and modulation matrix

- Use the **Mod on Daisy Editor 0.3.0** (Windows app, VST3 plug-in, or the browser page `editors/mod-on-daisy/web/index.html` in Chrome/Edge). Close other programs that hold the MIDI port.
- On connect the app **reads** the kit first; nothing is overwritten. The status line shows `streichwurst 0.1.0 · SQ + Matrix`.
- Pages **Strings, Solo, Tastatur (keyboard), FX, Global** hold every parameter. Pot moves on the kit appear in the app.
- **Matrix** (top) switches to the 8 × 8 modulation matrix; **Klangparameter** switches back. Sources: LFO 1/2, Random, CV 1/2, mod wheel, channel and poly aftertouch. Targets: Registration, Animate, Phaser, Reverb, Balance, Solo Tone, Tremolo (±100 %) and Pitch (±1 semitone). At most eight active connections; an empty matrix leaves the sound unchanged.
- **Patch speichern / laden** saves a `.swpatch` file with all parameters and the matrix. Loading shows what fits; only **Auf Gerät übertragen** sends it. In the VST3 plug-in the patch is stored in the song.
- **Zähler lesen** shows diagnostics, e.g. callback peak (CPU) and free stack.

> The app's interface is in German. The FX image shows only the FX page and no matrix.

7. The FX image (StreichwurstFx-v0.1.0.bin)

Line in → Animate, Ensemble, Phaser → Reverb → out. Plug only the left input for mono to both sides. The pots act only once they are moved.

Control	Function
VR1 – VR6	Animate, Animate rate, Phaser, Phaser rate, Reverb, Reverb decay
VR7	Input level (unity at about 70 % of the travel)
VR8	Output level
S1 / S2 / S3	Ensemble mode 0 / 1 / 2 (LED shows the mode)
S12	Ensemble on/off (LED)
S16	Clear delay line and reverb tail (on release); LED S16 = CPU overrun since start
CC 102 – 109	The FX parameters on any MIDI channel (USB and DIN)

With the ensemble off and Animate, Phaser and Reverb at 0 the signal passes unchanged.

8. Parameter reference

Instrument

Parameter	Range	Default	CC	Note
Registration	0 – 100 %	50 %	20	
Octaves	0 Base, 1 Both, 2 Upper	Base	21	
Crescendo	0 – 100 %	20 %	22	
Release (strings)	0 – 100 %	45 %	23	
Solo Tone	0 – 100 %	50 %	24	
Solo Attack	0 – 100 %	10 %	25	
Solo Release	0 – 100 %	30 %	26	
Solo sustain mode	0 Decay to level, 1 Hold, 2 Percussive	Decay to level	27	
Solo Sustain	0 – 100 %	70 %	28	
Tremolo depth	0 – 100 %	0 %	29	
Tremolo rate	0.5 – 12 Hz	5 Hz	30	log scale
Layer / Split	0 Layer, 1 Split	Layer	31	
Split key (MIDI note)	0 – 127	60	85	
Velocity (strings)	0 – 100 %	0 %	86	own addition
Velocity (solo)	0 – 100 %	100 %	87	own addition
Bend range	0 – 12 st	2 st	–	
Transpose	-24 – 24 st	0 st	–	
Tune	-50 – 50 cent	0 cent	–	own addition

Effects

Parameter	Range	Default	CC	Note
Animate	0 – 100 %	0 %	102	
Animate rate	0.05 – 5 Hz	0.5 Hz	103	log scale, own addition
Ensemble on/off	off / on	on	104	
Ensemble mode	0 Mode 0 (3 taps), 1 Mode 1 (dry + 5), 2 Mode 2 (dry + 3)	Mode 1 (dry + 5)	105	
Phaser	0 – 100 %	0 %	106	
Phaser rate	0.05 – 5 Hz	0.4 Hz	107	log scale, own addition
Reverb	0 – 100 %	20 %	108	
Reverb decay	0 – 100 %	50 %	109	own addition

Global

Parameter	Range	Default	CC	Note
Balance strings/solo	0 – 100 %	50 %	8	
Master	0 – 100 %	80 %	7	
MIDI channel (0 = omni)	0 – 16	0	–	
Pot page on the kit	0 Main, 1 Solo, 2 FX	Main	–	read only
RAM preset (last recalled)	0 – 3	0	–	read only

“Own addition” marks controls the original does not have as separate knobs.

9. Troubleshooting

Observation	Check
No sound from MIDI	MIDI channel on the Global page (0 = omni); USB port held by another program?
A pot does nothing	Soft takeover: turn it across the current value. Check the page (S9 LEDs).
Strings only, no solo (or the reverse)	Balance (VR4 on Main); Layer/Split and split key (S14, VR7 on Solo).
Volume jumps when switching the ensemble	Expected: the ensemble mode sets the overall level, as in the original.
Hanging notes	S16 (panic) or CC123; report the situation.
Crackles, drop-outs	Note the setting and read the counters in the app (callback peak, overruns); report them.
The app shows no Streichwurst pages	Editor 0.3.0 or newer is needed.

10. Test sheet

Record results with the date as **MEASURED**; anything not checked stays **NOT MEASURED**.

#	Step	Expected	Result
1	FX image: line source, defaults	widened chorus, some reverb	
2	FX image: S12 off, VR1/VR3/VR5 at 0	signal unchanged	
3	Instrument: power up, play S1–S8	strings in ensemble mode 1, stereo	
4	MIDI keyboard over USB and DIN, bend, pedal	notes, ±2 st bend, pedal holds	
5	Main page VR1–VR8	each control audible, no jumps	
6	S9 pages, S10–S15	pages and switches as in section 4	
7	Split (S14) with solo on the upper keys	strings below, solo above	
8	Dense chord, octaves Both, mode 1, reverb	no drop-outs; app counter overruns = 0	
9	App: connect, change values, matrix LFO 1 → Registration	changes audible, pots visible in the app	
10	S16 panic, S9 + S16 for 5 s	silence; bootloader with all LEDs	

Full test lists: `effects/streichwurst-fx/docs/HARDWARE_TEST.md` (F1–F14) and `instruments/streichwurst/docs/HARDWARE_TEST.md` (H1–H20).

Sources: instruments/streichwurst/docs (CONTROLS, PARAMETERS, FIDELITY, VALIDATION), effects/streichwurst-fx/README.md · Parameter tables generated from contracts/parameters.json · Repository sscheidl/Daisy-Seed, PR #77